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Non-Human Primate Recognition using Convolutional Neural Networks

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Abstract

Recent advancements in machine learning, coupled with the availability of affordable and powerful hardware, have led to a surge in studies within the interdisciplinary field of animal recognition using computer vision. These studies have enabled significant achievements such as monitoring invasive species, protecting against poaching, improving public road safety, estimating population sizes, and enhancing animal welfare both in captivity and in the wild. Despite these successes, a concise set of guidelines to help small-scale research projects achieve high accuracy in animal classification models remains lacking. This research provides a comprehensive review of the current state of the field, highlighting common challenges and solutions emerging from collaborations between animal experts and computer scientists. A subset of these solutions has been tested on publicly available non-human primate face datasets, successfully identifying individuals and broader species with high accuracy. The findings demonstrate that small, deliberate adjustments to convolutional neural network models during training can yield accuracy above 95%, even on small and unbalanced datasets. These conclusions are significant for the broader goal of animal monitoring, as they underscore the feasibility of integrating machine learning as a cost-effective and time-saving tool for processing large volumes of visual data.

Key words: convolutional neural networks, machine learning, animal recognition, primate recognition, image classification

Introduction

Due to the increase of knowledge in machine learning, animal recognition using computer vision has become a realistic and feasible tool for animal monitoring. Video recording equipment, both in wildlife camera traps and habitat enclosures, provides a safe, non-invasive way to observe animals with minimal human interaction (1). These types of cameras then provide large amounts of photos and videos, which requires hours of time and expense to analyze. Machine learning, specifically through convolutional neural networks, provides an efficient way to analyze this footage quickly. This technology has been used to estimate population sizes, prevent poaching, increase public safety, monitor invasive species, and many other animal research needs.

The aim of this research is to review the current state of the field in machine learning animal recognition systems and to explore common difficulties and solutions available to wildlife

experts and computer scientists who are conducting future interdisciplinary studies. Many of the convolutional neural network solutions have been applied and analyzed on public datasets for non-human primate facial recognition. To keep the information concise, this study has been limited to image classification (assigning a label to an image) and not object detection (finding a bounding box for a particular object in an image).

Related Work

Over the last fifteen years, researchers have been able to apply machine learning to many areas of animal research. These studies can be split into course-grained identification, which identifies a species from among multiple species (2; 3; 4), and fine-grained identification, which identifies individuals from a single species (5; 6; 7). Although similar in many regards, these

two types of studies each present unique difficulties which must be overcome.

For animals with identifiable markings, convolutional neural networks are able to learn the patterns of individuals in species such as tigers (8), zebras, jaguars (9), whales (10), penguins (11), and elephants (12). For animals without these distinct markings, inspiration has been drawn from the successful application of machine learning to human facial recognition first by the seminal work by Viola and Jones that used hand-engineered features (13), then later by convolutional neural network improvements as GPUs became more powerful and accessible (14). By isolating and training a model on the face region, researchers were able to achieve a top 1 accuracy of 96.27% for giant pandas (5) and 83.9% for brown bears (6).

Machine learning animal recognition systems have a broad spectrum of applications, spanning both research and practical domains. For instance, in the study of social dynamics, researchers have leveraged these systems to analyze cohabitation times within animal groups, which allows for the mapping of network structures and their evolution over time (15). Similarly, in more practical settings, such as traffic safety, kangaroo detection systems have been developed to enhance vehicle sensors, thereby reducing the risk of collisions (16).

Moreover, the efficient processing of camera trap images in wildlife monitoring has proven invaluable for various conservation efforts. One key application is in anti-poaching initiatives, where machine learning models can detect human presence in protected areas (17). Yousif et al., for example, demonstrated this by developing a CNN model capable of distinguishing between humans, animals, and the background in camera trap images (18). Finally, in the agricultural sector, machine learning networks have been employed to monitor invasive species, such as badgers on farmland, aiding in the protection of crops and local wildlife (19).

Primates

One of the most prevalent uses of non-human facial recognition has been in primates. Similar to the human recognition studies, early research in 2012-2013 by Loos and Ernst utilized Gabor features and SURF descriptors in conjunction with a support vector machine (SVM) for chimpanzee classification (20) achieving an accuracy of 82.88%. This presented an early step forward in the field, but the algorithm was highly-specialized for chimpanzees only, and its success was greatly dependent on face alignment to place the eyes, nose, and mouth in a similar pixel location before identification. This alignment process is shown in Figure 1. In future applications with CNNs, studies have either chosen to use image rotations to train the model on various head positions (15; 21; 22) or they have pre-processed the image to orient the head to an upright position (23; 5; 24; 25). Although both methods have proved successful to increase training accuracy, pre-processing the face orientation requires that significant features, such as both eyes and the nose, are visible, and they must be detected and annotated either manually or by using another detection network (24).

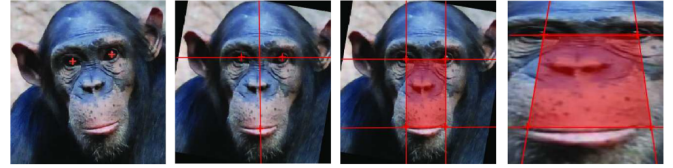


Fig. 1. Facial alignment before image classification (20)

Subsequent work by Loos et al. in 2016 used similar methods to detect chimpanzees in purely camera trap recordings (26). Using the same CZoo and CTai dataset from the 2013 studies, Freytag et al. in 2016 eliminated the need for pre-processing facial alignment and hand-engineering features by introducing convolutional neural networks (CNNs). This improved the accuracy to 91.99% (27). With the expansion of CNN architectures at this time and the precedence of success set by Loos and others in Germany, use of machine learning expanded into identification of many more primates such as rhesus macaques (24), Japanese macaques (22), bonobos (28), golden monkeys (29), lemurs (23), and gorillas (30).

Brief Evolution of Convolutional Neural Networks

A major catalyst for the development of convolutional neural network architectures has been the ImageNet Large Scale Visual Recognition Challenge (ILSVRC). The goal of this competition was to design an architecture that will obtain the highest accuracy on a subset of the ImageNet dataset, comprised of over 1.2 million images labeled with around 1,000 highly-diverse categories (31). In 2012, Krizhevsky et al. developed the AlexNet architecture which achieved state of the art results of a top-5 error rate of 17.0% using what would become fundamental techniques in machine learning: ReLU non-linearity, dropout (32), stochastic gradient descent (SGD), and data augmentation. These techniques can help prevent a model from overfitting, a problem that occurs when the model memorizes features from the training dataset and cannot obtain high accuracy on images that weren't part of training. The AlexNet architecture consisted of a shallow network of 5 convolutional layers and 3 dense, fully-connected layers as shown in Figure 2.

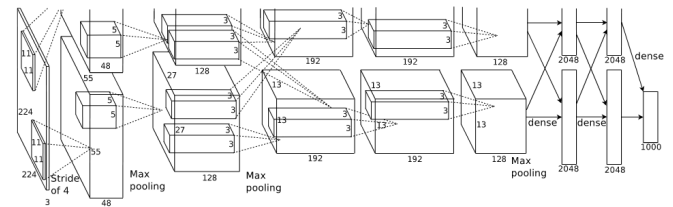


Fig. 2. AlexNet: Winning architecture from ILSVRC 2012 (31)

In 2014, Szegedy et al. at Google introduced GoogLeNet (also known as InceptionV1), a 22-layer architecture comprised of many blocks of Inception modules. The GoogLeNet architecture achieved a top-5 error rate of 6.67%, winning the 2014 competition (33). Each Inception module concatenated 1x1, 3x3, and 5x5 convolutions to capture a wide range of feature sizes at once, as shown in Figure 3. The Google team then evolved and fine-tuned this Inception concept into

InceptionV2, InceptionV3 (34), InceptionV4, and a Inception-ResNet hybrid architecture (35).

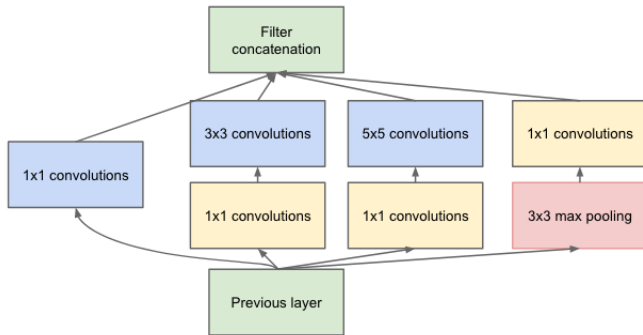


Fig. 3. Inception Module from the GoogleNet architecture (33)

In 2015, the ResNet architecture was introduced by He et al. at Microsoft Research (36). When trying to make the architecture deeper (with more convolutional layers), they ran into the common issues of gradients either exploding or vanishing as they were continually multiplied together. To solve this problem, the team introduced shortcut connections. Using an ensemble of architectures, the ResNet model achieved a 3.57% top-5 error rate and won the ILSVRC 2015 competition. Many variants of ResNet were released with various depths, 50, 101, 152, etc. As no one single architecture is ideal for every dataset, researchers have chosen many of these architectures for transfer learning such as VGG-M and ResNet-50 in Schofield et al. (15); ResNet-152 in Norouzzadeh et al. (37); ResNet-101, 50, and 18 in Chen et al. (5); and ResNet-18 and DenseNet-121 in Shukla et al. (21).

A wide range of CNN architectures have since been developed; a review of which would be beyond the scope of this research. It was these recent developments and availability of promising new architectures that have led to the prevalence and success of animal recognition systems. By leveraging these advanced machine learning architectures, creating an effective animal identification system has become more attainable than ever, even for small-scale research projects.

Common Challenges and Solutions

Although coarse-grain species identification and fine-grain individual identification problems will vary greatly between different animal species and datasets, there are a few notable difficulties addressed by each of the studies that have been able to successfully implement machine learning system. Outlined below are a few of these prevalent difficulties and ways to solve them.

Limited availability of large datasets

One of the substantial constraints in implementing a machine learning system for animal research is obtaining a suitable dataset containing thousands of subject images that have been expertly-labeled with either the species or individual name, dependent on the type of study being conducted.

- **Citizen Science** - For species studies, common animals can be identified by individuals who do not have formal

expertise; thus, citizen scientists can volunteer to help label large volumes of images. The Snapshot Serengeti project used motion-sensor camera traps in Serengeti National Park, Tanzania to record and label 2.65 million sequences of images among 61 categories (38). On this dataset, Norouzzadeh et al. achieved a 99.3% accuracy rate (37). They theorize that the weights learned from training this large model could be used in transfer learning to smaller camera-trap studies, potentially more than the standard transfer learning using ImageNet weights. The Zooniverse project is a platform that facilitates the connection between citizen scientists and dataset labeling projects. Popular projects include Snapshot Wisconsin which used over 21,000 volunteers to classify over 197,000 subjects from trail cameras (39). These projects allow researchers to more easily obtain large, labeled datasets with a wide variety of environments and species (3).

- **Image Labeling Software** - Annotating thousands of images with classifications and bounding box coordinates can be a time-consuming process. To assist this process, computer science researchers have developed software designed to support the creation of machine learning datasets. A recent publicly-available application is the open-source Timelapse application (40) which can be customized to meet the individual needs of camera-trap studies. The Visual Geometry Group at Oxford, who pioneered the VGGNet architecture and won the ILSVRC 2014 localisation task (41), released a free, online tool to annotate images, audio, and video (42).
- **Transfer Learning** - On projects containing limited images for CNN training, transfer learning is a way to increase accuracy by utilizing weights that have been previously-learned on large-scale datasets, such as ImageNet. In a study on the transferability of transfer learning, Yosinski et al. demonstrated that transfer learning was most effective when the pre-training dataset was similar to the final dataset (such as man-made or natural objects), but even dissimilar transfer learning datasets showed a marked improvement over random initialization of weights (43).
- **Synthetically-Generated Images** - Due to the improvements of simulated images from both video game rendering and artificial intelligence, it is becoming feasible to generate artificial images to supplement a lack of training data. This technique was used by Saleh et al. to generate over 17,000 images of kangaroos in urban traffic scenarios, as shown in Figure 4 (16).

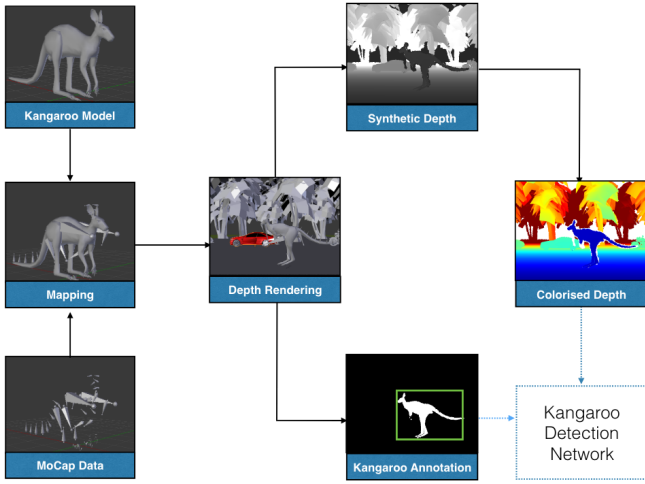


Fig. 4. Synthetic depth images of kangaroos in traffic scenarios (16).

Variation in pose and lighting conditions

Collecting high-quality images of individual animals can have unique challenges. Images taken at different times of day can produce varying brightness and color conditions. Unpredictable behavior and fast movements can result in difficulty obtaining images that show the face region facing towards the camera. To train an effective machine learning model, it is essential to account for these challenges both during and after image collection.

- **Image Collection Best Practices** - Through the collection of a significant number of images in difficult field conditions, researchers such as Guo et al. have outlined guidelines and best-practices for field practitioners collecting images for machine learning systems (29). They recommend systematically placing cameras facing opposite directions in populated areas. Before starting, simulation studies can be conducted to map an ideal location for data collection (26). It is recommended to obtain images that are not under strong light or shadows. Head rotation (on an axis that cannot be fixed in post-processing) is generally recommended to be between ± 15 degrees from a front-facing orientation (29).
- **Data Augmentation** - By adjusting image factors such as spatial and color transformations, this allows the dataset to be bolstered with additional images that can be used during model training. It also makes the training more robust to changes in lighting conditions and facial rotations. Common spatial transformations in animal studies include translation, rotation, cropping (27), and flipping on the horizontal or vertical axis (5; 22). Color transformations include adjustments to saturation, contrast, hue, brightness(15), and histogram equalization (20).

Datasets

CZoo Dataset - This dataset, originally created by Loos and Ernst (20), contains 2,109 cropped chimpanzee faces labeled with 24 individuals. The images are distributed fairly evenly between individuals, as shown in Figure 5 with an average of

87.88 images per individual. As it was one of the early publicly-available datasets, it has been used as a benchmark against which many other primate recognition systems have been tested (23; 21; 28; 27).

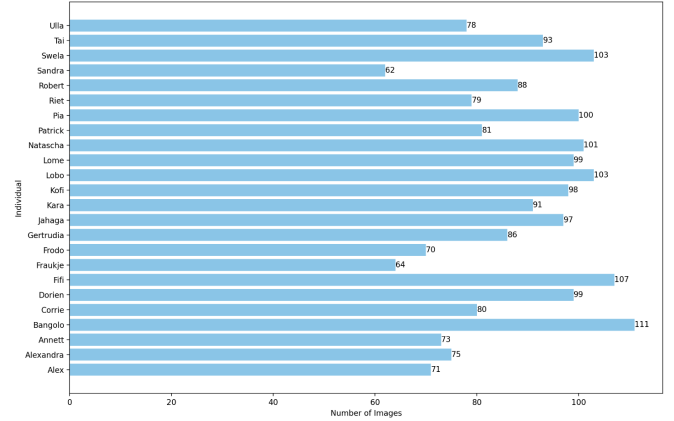


Fig. 5. Distribution of images in the CZoo Dataset

C'Tai Dataset - This dataset, also created by Loos and Ernst (20), contains 5,078 cropped chimpanzee faces labeled with 78 individuals. This dataset is noted for having significantly worse image quality and distribution of individuals, with some classes having as little as 1 image while others have 344. Additionally, the images display a large variance of lighting conditions, head orientations, and occlusions.

Animal Face Dataset (AFD) - The Animal Face Dataset, created by Guo et al. (29), has a publicly-available dataset that contains 12,599 images distributed between 17 non-human primate species. This is a subset of the larger, 102,399-image collection that must be obtained directly from the author. The public dataset distribution is highly-unbalanced towards *Rhinopithecus roxellanae* (Golden snub-nosed monkey, 34.46%) and *Macaca mulatta* (rhesus macaque, 31.85%).

Experiments

Technology used

All experiments were completed in Python using TensorFlow and Keras. The tensorflow-metal plugin (44) was used to utilize the Apple silicon GPU to accelerate training times. The device used for training was a 2023 Mac Mini containing an Apple M2 Pro chip with a 10-core CPU and 16-core Apple silicon GPU. Model training graphics were created with TensorBoard (45).

Image Augmentations

All images in the CZoo dataset were pre-processed to resize the images to 224x224 pixels. For each image, 5 augmentations were applied: horizontal flip, random rotation of $\pm 10\%$, random saturation, random contrast, and random hue, as shown in Figure 6. Parameters for the augmentations were evaluated by testing a series of parameter values on multiple images from the dataset. Values were chosen for ranges that had a visual impact on the image, without obscuring features on darker images from the set. With an ensemble of all augmentations, these adjustments increased the dataset from 2,109 images to 12,654.

Due to the very limited size of the dataset, each augmentation was completed on 100% of the images in the dataset, similar to the methodology used by Chen et al. (5).

Model training cycles were completed for 60 epochs on the C-Zoo dataset to evaluate the effect that each individual image augmentation had on the final outcome. Results for these tests are shown in Figure 7. The ensemble training cycle contained the full dataset with all five of the augmentations included. The saturation adjustment had the greatest improvement on overfitting, improving the validation accuracy by 44.45% from the non-augmented dataset. In the subsequent models being tested, the ensemble data augmentations were applied. Due to the increased validation accuracy of applying only the saturation adjustment, the final results shown in Table 1 use saturation as the only augmentation.

Transfer Learning Model Selection

Transfer learning is a common technique to improve model accuracy when limited training data is available. In many animal recognition studies, there is limited discussion on the model architecture chosen for transfer learning. As shown in Figure 8, seven transfer learning models were selected to test a variety of architectures. The models were frozen to not allow retraining of the weights on any layer. The final fully-connected layer was not included to allow for the model to be customized for the number of classes in the new dataset. The standard ImageNet weights were used for all transfer learning models. All tests had identical hyperparameters: Adam optimizer, learning rate of 0.001, batch size of 32, and 30 epochs. The CZoo dataset was randomly split into a 90/10 training and validation split with 11,389 images used for training and 1,265 images used for validation. In both training and validation accuracy, InceptionV3 outperformed the other architectures, with a training accuracy difference of over 10% above the second best architecture, DenseNet201. As a result of this experiment, the InceptionV3 architecture was chosen for the remaining training cycles.

Hyperparameter Grid Search

To fine-tune the hyperparameters, a grid search was performed to train a model on each combination of optimizer, learning rate, and batch size. This tested the optimizers: Adam, SGD, and RMSProp, with 0 momentum or decay. Learning rates were testing for 0.1, 0.01, 0.001, and 0.0001. Batch sizes were tested for 16, 32, and 64. This resulted in a significant range of values, with a difference between the best and worst hyperparameter combination being 66.54%. In future work, further trials could be conducted to test the effect of optimizer momentum and decay on these results.

Top Layers

After the transfer learning model was selected, eight model architectures were trained to vary the structure of layers included after the transfer learning model. These include flatten, dense, average pooling, dropout (32), and batch normalization (46). Results from these trials are shown in Figure 9.

Generalization to Other Datasets

Once the model was fine-tuned to produce improved results on the CZoo dataset, models were trained to see if the architecture would generalize well for other datasets. Top results for the

CZoo, CTai, and AFD datasets are shown in Table 1. Similar to the results produced when completing the augmentation training cycles, all datasets had a reduced overfitting of 5%-9% when using saturation as the only image augmentation.

Table 1. Generalization to multiple datasets

Dataset	Training Accuracy (%)	Validation Accuracy (%)
CZoo ¹	99.5	94.54
CTai ¹	95.46	81.2
AFD ²	99.33	97.82

¹Fine-grain (individual classification)

²Course-grain (species classification)

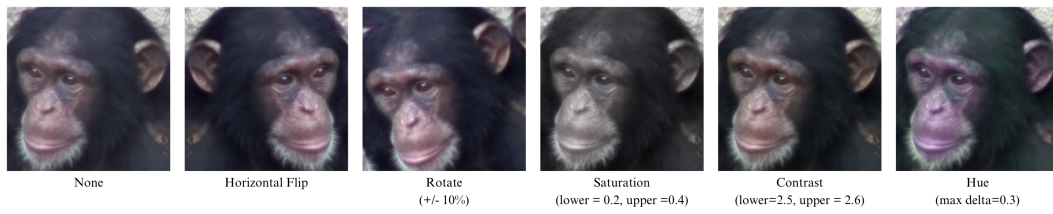


Fig. 6. Example of the image augmentations applied to the CZoo dataset.

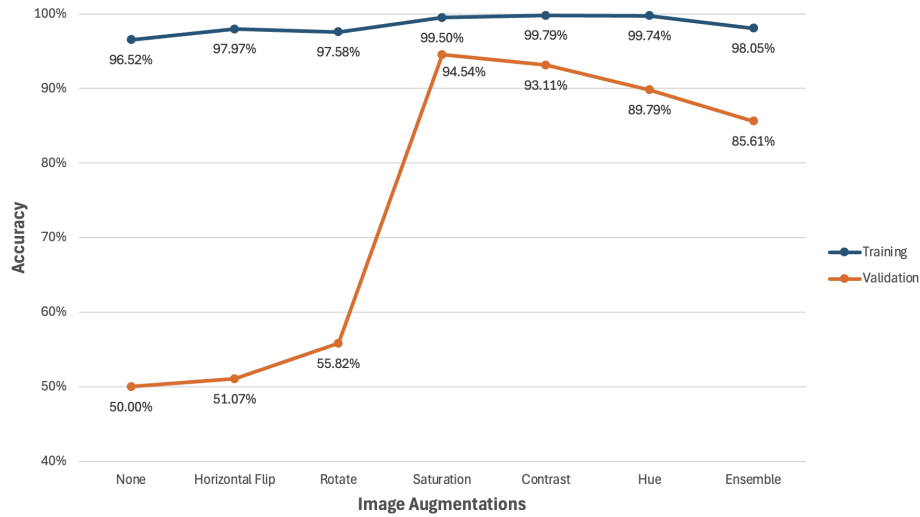


Fig. 7. Effect of individual and combined (ensemble) image augmentations on reducing overfitting.

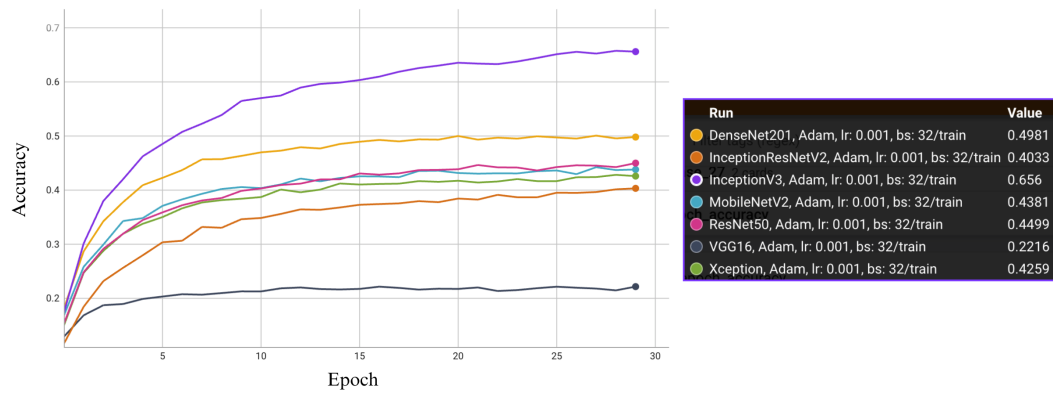


Fig. 8. Accuracy results of testing various architectures for transfer learning.

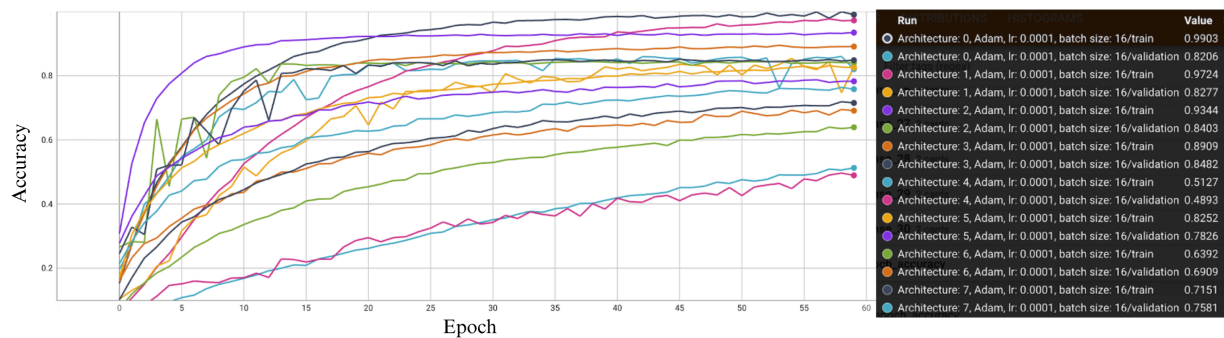


Fig. 9. Accuracy results of testing various model layers placed after InceptionV3 architecture.

Results and Future Work

By utilizing the proposed solutions to animal recognition difficulties, our research was able to achieve high training accuracy of 99.5% with minimal overfitting. This model also successfully generalized to distinct domains of primate recognition in both individual classification and species classification.

In future work, there is room to further develop the underutilized technique of transfer learning using pre-train weights from similar domains, as suggested by Norouzzadeh et al. (37). This has the potential to provide a more robust model that can adapt to challenging pose and lighting conditions that are common in camera-trap field work. Additional work can also be completed to analyze the results of momentum and decay in the optimization functions.

Conclusion

Recent enhancements in machine learning and computer vision have introduced a promising new method of animal monitoring. Interdisciplinary research between computer scientists and biological practitioners provides an opportunity to expand knowledge about convolutional neural networks, while also protecting public health and animal well-being (1; 47; 48). As it has been demonstrated in this study, by building off of the developments of complex CNN architectures, it is possible to achieve very promising results, even on small or limited datasets.

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Data availability

The code underlying this article is available at <https://github.com/KiyrahKeith/FaceIdentification>

Animal Face Dataset can be located at <https://data.mendeley.com/datasets/z3x59pv4bz/7>

CZoo and CTai Datasets can be located at https://github.com/cvjena/chimpanzee_faces/

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